

Emily Kopania – *Curriculum Vitae*

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Current Position

2023 – present NSF Postdoctoral Research Fellow
University of Pittsburgh
Advisor: Dr. Nathan Clark

Education

2016 – 2022 Organismal Biology and Ecology, Ph.D.
University of Montana
Advisor: Dr. Jeffrey Good

2015 – 2016 Molecular Genetics and Biochemistry, M.S.
University of Southern California
Advisor: Dr. Matthew Dean

2012 – 2016 Biological Sciences, B.S., *magna cum laude*
Computational Biology and Bioinformatics, minor
University of Southern California

Professional Appointments

2022 – 2023 Postdoctoral Research Associate
University of Utah
Advisor: Dr. Nathan Clark

2013 – 2015 Undergraduate Research Assistant
University of Southern California
Advisor: Dr. Matthew Dean

Manuscripts in Preparation

1. **Kopania, E. E. K.**, J. Denby*, and N. L. Clark. “Relative evolutionary rates of mammalian reproductive genes in response to sperm competition pressures.” *In prep.*
*Undergraduate co-author
2. Meyer, G. H., **E. E. K. Kopania**, M. Tene, A. Kowalczyk, R. Redlich, A. R. Pfenning, W. K. Meyer, M. Chikina, N. L. Clark. “RERconverge update: Drastic runtime reduction and correlation overhaul.” *Submitted.*

Peer-Reviewed Publications

12. Clark, N. L., A. Kowalczyk, **E. E. K. Kopania**, and M. Chikina. (2025). “Phylogenomic approaches to study adaptive evolution in mammals: From aging to aquatic lifestyles.” *Annu Rev Genet.* 59:461-483.

11. **Kopania, E. E. K.** and N. L. Clark. (2025). "Mammalian retinal specializations for high acuity vision evolve in response to both foraging strategies and morphological constraints." *Evol Lett.* qrae072.
10. **Kopania, E. E. K.**, G. W. C. Thomas, C. R. Hutter, S. M. E. Mortimer, C. M. Callahan, E. Roycroft, A. S. Achmadi, W. G. Breed, N. L. Clark, J. A. Esselstyn, K. C. Rowe, J. M. Good. (2025). "Sperm competition intensity shapes divergence in both sperm morphology and reproductive genes across murine rodents." *Evolution.* 79(1):qpae146.
Editor's Choice article in the January 2025 issue of *Evolution*
9. Kang, T., E. C. Moore, **E. E. K. Kopania**, C. D. King, B. Schilling, J. Campisi, J. M. Good, R. B. Brem. (2023). "A natural variation-based screen in mouse cells reveals USF2 as a regulator of the DNA damage response and cellular senescence." *G3 Genes Genom Genet.* 13(7):jkad091.
8. **Kopania, E. E. K.**, E. M. Watson, C. C. Rathje, B. M. Skinner, P. J. I. Ellis, E. L. Larson, J. M. Good. (2022). "The contribution of sex chromosome conflict to disrupted spermatogenesis in hybrid house mice." *Genetics.* 222(4):iyac151.
7. Moore, E. C., G. W. C. Thomas, S. Mortimer, **E. E. K. Kopania**, K. E. Hunnicutt, Z. J. Clare-Salzler, E. L. Larson, J. M. Good. (2022). "The evolution of widespread recombination suppression on the Dwarf Hamster (*Phodopus*) X chromosome." *Genome Biol Evol.* 14(6):evac080.
6. **Kopania, E. E. K.**, E. L. Larson, C. Callahan, S. Keeble and J. M. Good. (2022). "Molecular evolution across mouse spermatogenesis." *Mol Biol Evol.* 39(2):msac023.
5. Larson, E. L., **E. E. K. Kopania**, K. E. Hunnicutt, D. Vanderpool, S. Keeble, J. M. Good. (2021). "Stage-specific disruption of X chromosome expression during spermatogenesis in sterile house mouse hybrids." *G3 Genes Genom Genet.* 12(2):jkab407.
4. Skinner, B. M., C. C. Rathje, J. Bacon, E. E. P. Johnson, E. L. Larson, **E. E. K. Kopania**, J. M. Good, G. Yousafzai, N. A. Affara, and P. J. I. Ellis. (2019). "A high-throughput method for unbiased quantitation and categorisation of nuclear morphology." *Biol Reprod.* 100(5):1250–1260.
3. Skinner, B. M., J. Bacon, C. C. Rathje, E. L. Larson, **E. E. K. Kopania**, J. M. Good, N. A. Affara, and P. J. I. Ellis (2019). "Automated nuclear cartography reveals conserved sperm chromosome territory localization across 2 million years of mouse evolution." *Genes* 10(2): 109.
2. Larson, E. L., **E. E. K. Kopania**, and J. M. Good (2018). "Spermatogenesis and the evolution of mammalian sex chromosomes." *Trends Genet* 34(9):722-732.
1. Chang, P. L., **E. Kopania**, S. Keeble, B. Sarver, E. Larson, A. Orth, K. Belkhir, P. Boursot, F. Bonhomme, J. M. Good, and M. D. Dean (2017). "Whole exome sequencing of wild-derived inbred strains of mice improves power to link phenotype and genotype." *Mamm Genome* 28(9-10):416-425.

Grants and Fellowships

2025	University of Pittsburgh Center for Evolutionary Biology and Medicine Catalyst Outreach Award	\$10,000
2023	National Science Foundation Postdoctoral Research Fellowship in Biology	\$240,000
2023	National Institutes of Health NRSA Ruth Kirschstein Postdoctoral Fellowship (F32)	Declined
2020	University of Montana Dr. Linda Phillips Knoblock Fellowship	\$7000
2020	University of Montana Drollinger-Dial Travel Award	\$1000

2017 – 2020	National Science Foundation Graduate Research Fellow	\$138,000
2017	Society for the Study of Evolution Rosemary Grant Award	\$2500
2017	Society for Molecular Biology and Evolution	\$1500
	Young Investigator Travel Award	
2014, 2015	Rose Hills Undergraduate Research Fellowship	\$10,000
2014	USC Provost Undergraduate Research Fellowship	\$3000

Honors and Awards

2025	University of Wisconsin Evolution Early Career Award Seminar
2016	Phi Beta Kappa Honor Society
2016	USC Discovery Scholar Award for Research
2015	Rose Hills Scholarship
2015	Richard M. and Patricia J. Mialovich Scholarship
2014	Albert Fisher Scholarship in Science
2013	USC Academic Achievement Award
2012 – 2016	USC Trojan Scholars Society - Presidential Scholar
2012 – 2016	National Merit Scholar
2012 – 2016	Sylas and Rose Marx Meyer Scholarship

Presentations

Invited presentations and presentations selected for symposia

1. **Kopania, E.** Oral presentation. University of Nevada Las Vegas School of Life Sciences. Las Vegas, Nevada, April 29, 2026.
2. **Kopania, E.** Oral presentation. City College of New York Department of Biology. New York, New York, March 12, 2026.
3. **Kopania, E.** Oral presentation (virtual). State University of New York Old Westbury Department of Biological Sciences. February 25, 2026.
4. **Kopania, E.** Oral presentation. Florida State University Department of Biological Science. Tallahassee, Florida, February 17, 2026.
5. **Kopania, E.** Oral presentation. Texas A&M University Department of Biology. College Station, Texas, February 9, 2026.
6. **Kopania, E.** Oral presentation. University of South Carolina Department of Biological Sciences. Columbia, South Carolina, January 14, 2026.
7. **Kopania, E., N. Clark.** "The evolution of high visual acuity specializations in mammals." Symposium Oral Presentation. 3rd Joint Congress on Evolutionary Biology. Montreal, Canada, July 26-30, 2024.
8. **Kopania, E.** "The phenotypic and molecular evolution of male reproduction across diverse murine rodents." Oral presentation. University of Denver Biology Seminar Series. Denver, Colorado, April 17, 2023.

National or international conferences

1. **Kopania, E., N. Clark.** "Evolution of gene expression in the trout retina." Oral presentation. Evolution Conference. Athens, Georgia. June 20-24, 2025.
2. **Kopania, E., N. Clark.** "Ecological pressures acting on mammalian visual traits." Poster. The Allied Genetics Conference. Washington, D.C., March 6-10, 2024.
3. **Kopania, E., G. Thomas, C. Hutter, S. Mortimer et al.** "Recurrent shifts in sperm competition intensity shape the phenotypic and molecular evolution of male reproduction across diverse

murine rodents." Poster. Society for Molecular Biology and Evolution Annual Meeting. Ferrara, Italy, July 23-27, 2023.

4. **Kopania, E.**, G. Thomas, C. Hutter, S. Mortimer et al. "Rodents of Unusual Sperm: Molecular and Phenotypic Evolution of Male Reproduction in Murine Rodents." Poster. Population, Evolutionary, and Quantitative Genetics Conference. Pacific Grove, California, June 7-10, 2022.
5. **Kopania, E. K.**, Larson, E. L., Good, J. M. "The molecular evolution of spermatogenesis." Poster. Biology of Spermatozoa Conference. Nynäshamn, Sweden, September 9-13, 2019.
6. **Kopania, E. K.**, Larson, E. L., Good, J. M. "The molecular evolution of spermatogenesis." Poster. Population, Evolutionary, and Quantitative Genetics Conference. Madison, Wisconsin, May 13-16, 2018.
7. **Kopania, E. K.**, Larson, E. L., Good, J. M. "Genomic conflict and the evolution of gene copy number in mice." Poster. Society for Molecular Biology and Evolution Annual Meeting. Austin, Texas, July 2-6, 2017.

Local conferences or seminar series

1. **Kopania, E.**, N. Clark. "Evolution of the retina in mammals and fishes." Oral Presentation. Pittsburgh Center for Evolutionary Biology and Medicine annual retreat. Bedford Springs, Pennsylvania, May 5-6, 2025.
2. **Kopania, E.**, N. Clark. "Mammalian retinal specializations are a main evolutionary response to ecological selective pressures on vision." Oral Presentation. University of Pittsburgh Molecular Evolution Laboratory Discussion. Pittsburgh, Pennsylvania, April 18, 2024.
3. **Kopania, E.**, N. Clark. "The evolution of teleost retinal specializations." Poster. Sensorium. Chicago, Illinois, November 11-12, 2023.
4. **Kopania, E.**, G. Thomas, C. Hutter, S. Mortimer et al. "Rodents of Unusual Sperm: Molecular and Phenotypic Evolution of Male Reproduction in Murine Rodents." Oral presentation. University of Utah Evolutionary Genetics and Genomics Seminar Series. Salt Lake City, Utah, November 2, 2022.
5. **Kopania, E. K.** "Molecular evolution of spermatogenesis in murine rodents." Oral presentation, public dissertation defense. University of Montana Ecology and Evolution Seminar Series. Missoula, Montana, January 19, 2022.
6. **Kopania, E. K.** "Genomic conflict and mouse sex chromosome evolution." Oral presentation. University of Montana Organismal Biology, Ecology, and Evolution Noon Seminar Series. Missoula, Montana, April 7, 2021.
7. **Kopania, E. K.** "Gene expression evolution across spermatogenesis." Oral presentation. University of Montana Organismal Biology, Ecology, and Evolution Noon Seminar Series. Missoula, Montana, October 2, 2019.
8. **Kopania, E. K.**, Larson, E. L., Good, J. M. "The molecular evolution of spermatogenesis." Poster. Symposium on the Evolutionary Genomics of Adaptations. Polson, Montana, June 1-3, 2018.
9. **Kopania, E. K.** "Mapping the genes responsible for variation in *Crassostrea gigas* shell morphology." Poster. University of Southern California Undergraduate Symposium for Scholarly and Creative Work. Los Angeles, California, April 15, 2015.

Teaching

Guest Lecturer

2026

BIOSC 1130

University of Pittsburgh

Evolution

2024	BIOSC 1540	University of Pittsburgh	Computational Biology
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Teaching Assistant

2021	BIOB 171	University of Montana	Principles of Biological Diversity
2017	BIOB 272	University of Montana	Genetics and Evolution
2016	BIOB 101	University of Montana	Discovering Biology

Other teaching experience

2015 – 2016	Supplemental Instruction Leader	University of Southern California
2015	Student Athlete Tutor	University of Southern California

Mentorship

Student-led projects

2024 – present	Justine Denby University of Pittsburgh honors undergraduate	Project: Molecular evolution of reproductive genes in response to sperm competition in mammals
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Co-mentored students

2023 – 2025	Alex Preble, Nico Schwartz, Krista Olson University of Pittsburgh undergraduates
2016 – 2021	Vanessa Knox, Stephenie Horne, Marc Velling, Sarah Wells, Dawit Mengistu University of Montana undergraduates

Professional Service and Development

Public engagement and education

2025	Participated in the McClintock Letter's initiative to write an Op Ed on the importance of science funding
2023	Salt Lake City School District Science & Engineering Fair judge
2018 – 2019	Montana Science Fair judge
2017 – 2019	University of Montana EmPower Place volunteer Led interactive science education activities at a learning center at the Missoula Food Bank
2017 – 2019	SciNation volunteer Led interactive science education activities at events on the Flathead Indian Reservation
2017	University of Montana SpectrUM Discovery Center volunteer Led interactive science education activities at a children's museum in Missoula, Montana
2017	We Are Montana in the Classroom Led interactive science education activities at events at elementary schools in rural Montana
2015 – 2016	USC Science Outreach (SCout) session leader Led weekly science lessons and experiments in 2nd and 3rd grade classes in Los Angeles public schools
2012 – 2013	QuikSCience mentor Mentored middle school students through a year-long research and conservation project

Diversity, equity, and inclusion (DEI) initiatives

- 2025 Field Safety Manual
University of Pittsburgh Biology Department
- Contributed to an update of my department's [fieldwork safety manual](#), which emphasizes inclusivity and safety concerns for field researchers from marginalized backgrounds
- 2022 – 2023 Equity, Diversity, and Inclusion Committee
University of Utah Department of Human Genetics
- Led department-wide DEI initiatives such as organizing DEI training at our annual retreat and changing department policy documents to use gender-inclusive language
- 2016 – 2022 Women in Science
University of Montana
- Organized networking and professional development events for women and other underrepresented groups in science

Other Professional service

- 2025 – present University of Pittsburgh Biology Postdoc Organization President
- Lead professional development and networking initiatives for postdocs, such as running an academic jobs application workshop
- 2024 – 2025 University of Pittsburgh Biology Postdoc Organization board member
- 2020 – present Peer reviewer: *Nature Communications*, *Molecular Biology and Evolution*, *PLOS Genetics*, *Proceedings B*, *Genome Biology and Evolution*, *Evolution*, *Molecular Ecology* (3 manuscripts), *PeerJ*, *Records of the Australian Museum*, *BMC Genomics*
- 2025 Postdoc fellowship applications mentor
Evolution 2025 Conference
- 2024 Postdoc fellowship applications mentor
3rd Joint Congress on Evolutionary Biology
- 2024 Guest editor, *Genome Biology and Evolution*
“Genomics of Adaptations to Extreme Environments” special section
- 2023 Symposium co-organizer, Society for Molecular Biology and Evolution
“Genomics of Adaptations to Extreme Environments”
- 2020 – 2021 Graduate student co-president and faculty liaison
University of Montana, ecology and evolution graduate program
- Organized networking, professional development, and DEI initiatives in my graduate program
 - Communicated graduate student ideas and concerns to faculty
- 2018 – 2019 Graduate and professional student association representative
University of Montana, ecology and evolution graduate program
- 2017 – 2018 Graduate student super speaker committee
University of Montana, ecology and evolution graduate program
- Organized a student-led visiting seminar speaker visit
- 2017 Student evaluation committee
University of Montana, Division of Biological Science
- Organized and summarized teaching evaluations

Professional Development

- 2025 Marine Biological Laboratory Visual Neuroscience Course, Woods Hole
- 2023 Allyship training, Society for Molecular Biology and Evolution

2020
2018

LGBTQ+ allies training, University of Montana
Evidence-based teaching in science course, University of Montana